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Abstract

The aim of the present thesis is to develop a modern machine learning architecture for lattice gauge theories. Gauge equivariant convolutional neural networks have shown that enforcing exact lattice gauge symmetry in a neural network significantly improves regression accuracy on gauge invariant observables. However, their convolutional kernels stay fixed after training and cannot adapt to configuration-dependent features. We introduce GELT (Gauge Equivariant Lattice Transformer), a gauge equivariant attention-based encoder network for lattice gauge theories that exploits the configuration-dependent attention mechanism to focus on the most physically relevant input features.

We test GELT on various regression targets involving physical observables and compare its performance against existing gauge-equivariant architectures. We also investigate the interpretability of the attention filters, studying whether attention localizes in topologically rich regions and whether the attention range correlates with physical correlation length.

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Chapter 1

Introduction

[TODO: Introduction: motivation, related work, contribution, thesis outline.]

Chapter 2

TODO: Background

[TODO: Background: theory and prior work needed to follow the thesis.]

Chapter 3

TODO: Methods

[TODO: Methods: the approach/architecture developed in this thesis.]

Chapter 4

TODO: Results

[TODO: Results: experiments, figures, and their discussion.]

Chapter 5

Conclusions

[TODO: Conclusions: summary of contributions, limitations, future work.]

Appendix A

TODO: Appendix

[TODO: Appendix content (e.g. notation, derivations, extra results).]

